

Program 10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

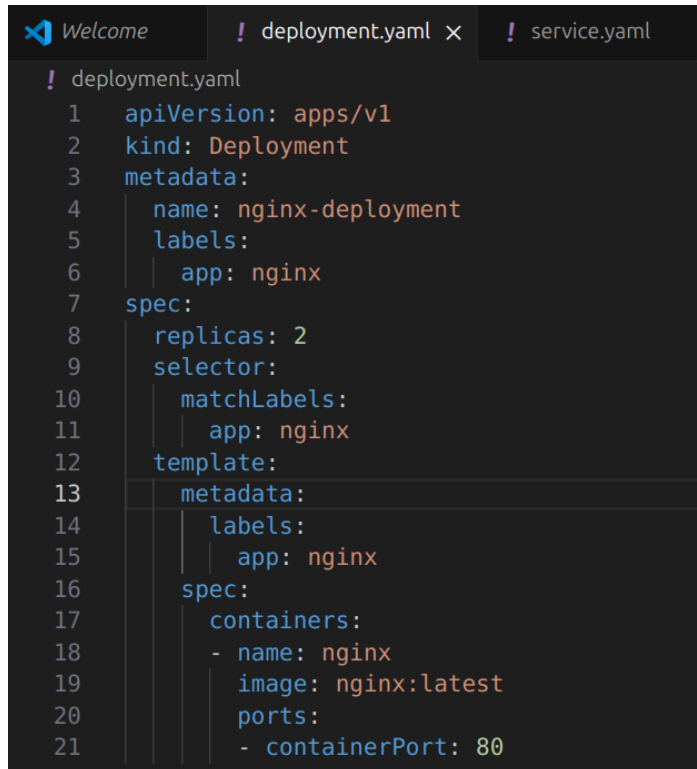
Step 1: Initialize Minikube

```
minikube start --cpus=2 --memory=4096
```

Step 2: Enable Kubernetes Dashboard

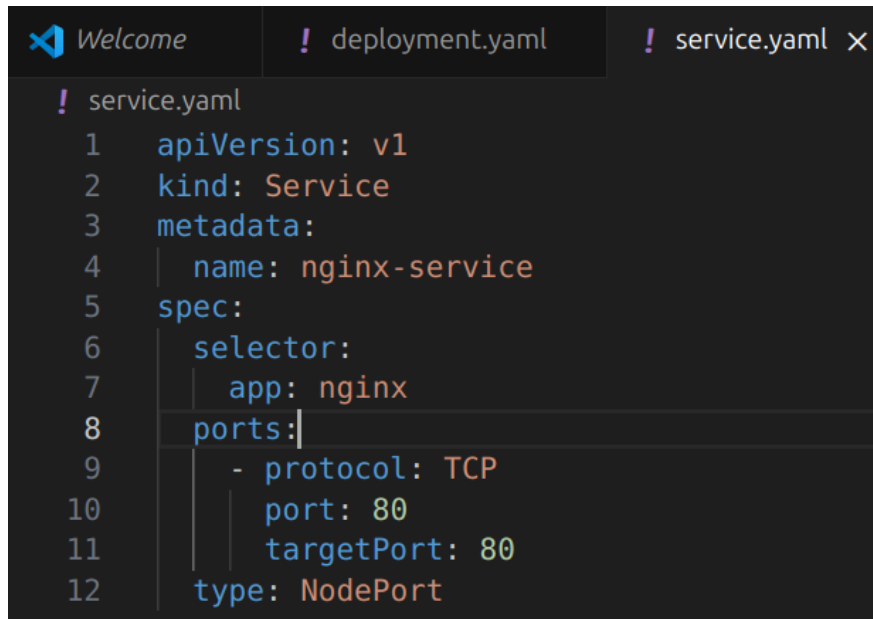
```
minikube dashboard
```

Step 3: Create the Deployment File

A screenshot of a code editor with three tabs: 'Welcome', 'deployment.yaml', and 'service.yaml'. The 'deployment.yaml' tab is active, showing a YAML configuration for a deployment. The code is as follows:

```
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5    labels:
6      app: nginx
7  spec:
8    replicas: 2
9    selector:
10     matchLabels:
11       app: nginx
12   template:
13     metadata:
14       labels:
15         app: nginx
16     spec:
17       containers:
18       - name: nginx
19         image: nginx:latest
20         ports:
21         - containerPort: 80
```

Step 4: Create the Service File

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'Welcome' with a blue icon, 'deployment.yaml' with a red exclamation mark icon, and 'service.yaml' with a red exclamation mark icon and a close button. The 'service.yaml' tab is active, showing a YAML configuration for a Service. The code is as follows:

```
! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: nginx-service
5  spec:
6    selector:
7      app: nginx
8    ports:
9      - protocol: TCP
10        port: 80
11        targetPort: 80
12    type: NodePort
```

Step 5: Apply Resources

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

Step 6 – 10: Monitor via CLI

```

rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_Lab/Lab10$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
mycron-29453251-mvs76               0/1     Completed 0           46m
mycron-29453252-lwwbg               0/1     Completed 0           45m
mycron-29453296-tcb76               0/1     Completed 0           24s
mycron-29453297-cq892               0/1     ContainerCreating 0           9s
nginx-deployment-6f9664446b-88j6b   1/1     Running   1 (44m ago) 75m
nginx-deployment-6f9664446b-nssd5   1/1     Running   1 (44m ago) 75m
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_Lab/Lab10$ kubectl get nodes
NAME      STATUS    ROLES    AGE   VERSION
minikube  Ready     control-plane  36d   v1.34.0
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_Lab/Lab10$ kubectl get svc
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes  ClusterIP   10.96.0.1     <none>         443/TCP    77m
nginx-service  NodePort    10.109.22.119 <none>         80:30301/TCP 76m
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_Lab/Lab10$ kubectl logs nginx-deployment-6f9664446b-88j6b
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/31 16:17:00 [notice] 1#1: using the "epoll" event method
2025/12/31 16:17:00 [notice] 1#1: nginx/1.29.4
2025/12/31 16:17:00 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/31 16:17:00 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/31 16:17:00 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/31 16:17:00 [notice] 1#1: start worker processes
2025/12/31 16:17:00 [notice] 1#1: start worker process 29
2025/12/31 16:17:00 [notice] 1#1: start worker process 30
2025/12/31 16:17:00 [notice] 1#1: start worker process 31
2025/12/31 16:17:00 [notice] 1#1: start worker process 32
2025/12/31 16:17:00 [notice] 1#1: start worker process 33
2025/12/31 16:17:00 [notice] 1#1: start worker process 34
2025/12/31 16:17:00 [notice] 1#1: start worker process 35
2025/12/31 16:17:00 [notice] 1#1: start worker process 36
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_Lab/Lab10$

```

Step 11: Access the Application

minikube service nginx-service

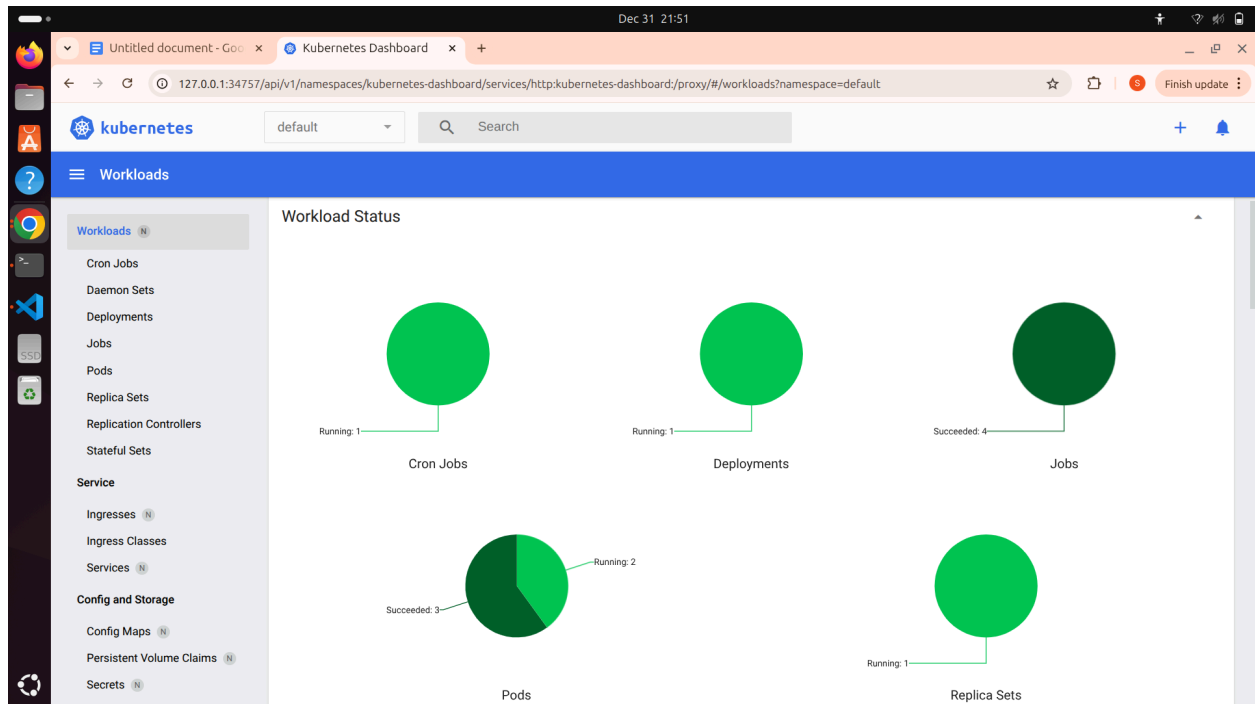
←
→
↺
⚠ Not secure
192.168.49.2:30301

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Step 12 – 14: Clean-up Process

`kubectl delete pods --all`

`kubectl delete svc --all`

`kubectl delete deployments --all`

```
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete pods --all
pod "mycron-29453300-d579v" deleted
pod "mycron-29453301-vg4rp" deleted
pod "mycron-29453302-f6bdf" deleted
pod "nginx-deployment-6f9664446b-88j6b" deleted
pod "nginx-deployment-6f9664446b-nssd5" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$
```